ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/325420416

A Short Method of Finding the Dual Basis of a Basis in a Finite Dimensional Vector Space.

Preprint · May 2018

DOI: 10.13140/RG.2.2.19526.37445/2

CITATIONS
0

READS
1,905

1 author:

Onurcan Bektaş
Middle East Technical University

3 PUBLICATIONS **0** CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Project

An Algebraic Method for the Application of the Constructive Proof of Classification Theorem for Closed and Connected Surfaces View project

All content following this page was uploaded by Onurcan Bektaş on 14 May 2019.

The user has requested enhancement of the downloaded file.

A Short Method of Finding the Dual Basis of a Basis in a Finite Dimensional Vector Space

Onurcan Bektaş*

Department of Mathematics
Department of Physics
Middle East Technical University
Ankara, Turkey

Dated: 30 May 2018 & Updated: May 14, 2019

The general problem:

Let V be a vector space of finite dimension over the field $\mathbb{F}$, where $\mathbb{F}$ can be any field, and $\beta' = \{v_1, ..., v_n\}$ be a given basis of V. We would like to compute the action of the dual basis $\alpha = \{f_1, ..., f_n\}$ of β' on a vector which is given in terms of a basis β.

The motivation for the problem arises from a special case:

When we work over $V = \mathbb{F}^n$, and want to compute the action of α to a vector which is given in terms of the standard basis, namely $\beta = \{(1, 0, ..., 0), (0, 1, 0, ..., 0)...(0, ..., 0, 1)\}$ of $V = \mathbb{F}^n$.The reason behind for this is to work with the standard basis is, in general, much easier.

Note that, by definition of the dual basis, we already know the action of α on a vector given in terms of β' since

$$f_i(v_j) = \delta_i^j \quad \forall v_i \in \beta', \quad \forall f_j \in \alpha.$$

Finding the action of f_is on a vector given in terms of β amounts to finding the matrix representations of f_is with respect to β.

Notations:

1-) $1s$ and $0s$ are going to denote the multiplicative and additive identities of the field $\mathbb{F}$.
2-) $\beta = \{e_1, ..., e_n\}$ is the standard basis of $\mathbb{F}^n$.
3-) $\begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ . \\ x_n \end{bmatrix}_\beta := x_1e_1 + x_2e_2 + ... + x_ne_n$ where $x_i \in \mathbb{F} \quad \forall i = 1, .., n.$
4-) $f = \begin{bmatrix} x_1 & x_2 & . & . & . & x_n \end{bmatrix}_\beta$ denotes that $f(e_i) = x_i \in \mathbb{F} \quad \forall i = 1, .., n.$
5-) $[g]|_\beta^{\beta'}$ denotes the matrix representation the linear map $g : V \to V$ where the input vector is given in terms of basis vectors in β and the image vector is given in terms of the basis vector in β'.

A special case: $V = \mathbb{F}^n$

Observe that, for any $f_i \in \alpha$, the matrix representation of f_i with respect to the basis β' is, by definition, $[0 \quad .. \quad 1 \quad .. \quad 0]$ where the only nonzero entry is in the $i - th$ place.

Since we know the action of f_is in terms of v_is -the basis elements in β' - we can find a map that transforms a vector given in terms of e_is to a vector given in terms of $v_i s$ without changing the vector itself, by composing that map with $f_i s$ from the right, we can find the action of f_is in terms of the basis elements e_is.

Let us take the identity operator $i : V \to V$, and write its matrix representation with respect to β in the domain and β' in the codomain, denoted as $[i]|_\beta^{\beta'}$. Since the identity operator does not change the input vector, and by definition of $[i]|_\beta^{\beta'}$ the image vector, i.e the output, is written in terms of the basis vectors in β'; this matrix does what we want. It transforms any vector given in terms of e_is to a vector given in terms of v_i

*Electronic address: `onurcan.bektas@metu.edu.tr`; Website: `users.metu.edu.tr/e214070/`

without actually changing the vector itself.

Now, the problem is reduced only to finding $[i]|_{\beta}^{\beta'}$. However, in general, this is hard to compute. That is why instead of doing the hard computation, we are going to find $[i]|_{\beta'}^{\beta}$, and then compute the inverse of $[i]|_{\beta'}^{\beta}$; the reason for this is because $([i]|_{\beta'}^{\beta})^{-1} = [i]|_{\beta}^{\beta'}$.

An example: Let $V = \mathbb{R}^n$, and $\beta' = \{(1,0,0),(1,1,0),(1,1,1)\}$, and lets find the action of α on a vector (x,y,z).
As usual, we know that

$$f_i = \begin{bmatrix} 0 & .. & 0 & 1 & 0 & .. & 1 \end{bmatrix}_{\beta'},$$

and,

$$id = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}_{\beta'}^{\beta}.$$

Note that, when we want to compute the action of id to, say $v_2 = (1,1,0)$, the matrix representation of v_2 in terms of β' becomes $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}_{\beta'}$.

Now, by ordinary methods employed in linear algebra, one can easily find the inverse of this matrix as

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}_{\beta}^{\beta'},$$

so that

$$[f_1] \cdot [i]_{\beta}^{\beta'} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}_{\beta'} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}_{\beta}^{\beta'} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}_{\beta}$$

$$[f_2] \cdot [i]_{\beta}^{\beta'} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}_{\beta'} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}_{\beta}^{\beta'} = \begin{bmatrix} 0 & 1 & -1 \end{bmatrix}_{\beta}$$

$$[f_3] \cdot [i]_{\beta}^{\beta'} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}_{\beta'} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}_{\beta}^{\beta'} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}_{\beta}$$

.

This means, in particular, that

$$f_1(xe_1 + ye_2 + ze_3) = x,$$
$$f_2(xe_1 + ye_2 + ze_3) = y - z,$$
$$f_3(xe_1 + ye_2 + ze_3) = z.$$

An application:

Using this method, one can easily compute the action of any linear map $g : V \to V$, whose action is known in terms of β', in terms of the β.

Since by our hypothesis, we know the action of g in terms of β', we know the matrix $[g]_{\beta'}^{\beta'}$. By the same argument used above, $[g]_{\beta'}^{\beta'}[id]_{\beta}^{\beta'}$ computes the action of g in terms of β, and $[id]_{\beta'}^{\beta}[g]_{\beta'}^{\beta'}[id]_{\beta}^{\beta'}$ computes the action of g in terms of β and outputs the result in terms of β also.

In the following updates of this paper, we are going to try to give a concrete method for the general case.

View publication stats